

Dietary intake of fruits and vegetables improves microvascular function in hypertensive subjects in a dose-dependent manner.

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BACKGROUND: Observational evidence has consistently linked increased fruit and vegetable consumption with reduced cardiovascular morbidity; however, there is little direct trial evidence to support the concept that fruit and vegetable consumption improves vascular function. This study assessed the dose-dependent effects of a fruit and vegetable intervention on arterial health in subjects with hypertension. **METHODS AND RESULTS:** After a 4-week run-in period during which fruit and vegetable intake was limited to 1 portion per day, participants were randomized to consume either 1, 3, or 6 portions daily for the next 8 weeks. Endothelium-dependent and -independent arterial vasodilator responses were assessed by venous occlusion plethysmography in the brachial circulation before and after intervention. Compliance was monitored with serial contemporaneous 4-day food records and by measuring concentrations of circulating dietary biomarkers. A total of 117 volunteers completed the 12-week study. Participants in the 1-, 3-, and 6-portions/d groups reported consuming on average 1.1, 3.2, and 5.6 portions of fruit and vegetables, respectively, and serum concentrations of lutein and beta-cryptoxanthin increased across the groups in a dose-dependent manner. For each 1-portion increase in reported fruit and vegetable consumption, there was a 6.2% improvement in forearm blood flow responses to intra-arterial administration of the endothelium-dependent vasodilator acetylcholine ($P=0.03$). There was no association between increased fruit and vegetable consumption and vasodilator responses to sodium nitroprusside, an endothelium-independent vasodilator. **CONCLUSIONS:** The present study illustrates that among hypertensive volunteers, increased fruit and vegetable consumption produces significant improvements in an established marker of endothelial function and cardiovascular prognosis.